A34312 No Calculator

UNIVERSITY OF BIRMINGHAM

School of Mathematics

Programmes in the School of Mathematics	Second Examination
Programmes in the School of Mathematics	Final Examination
Programmes involving Mathematics	Second Examination
Programmes involving Mathematics	Final Examination

2MVA 06 25667 Level I

LI Multivariable & Vector Analysis

2MVA3 06 35172 Level H

LH Multivariable & Vector Analysis

May/June Examinations 2023-24

Three Hours

Full marks will be obtained with complete answers to all FOUR questions. Each question carries equal weight. You are advised to initially spend no more than 45 minutes on each question and then to return to any incomplete questions if you have time at the end. An indication of the number of marks allocated to parts of questions is shown in square brackets.

No calculator is permitted in this examination.

Section A

1. (a) Consider the function

$$f(x,y,z) = e^{xy^2z^3}.$$

Calculate

(i) The direction and rate of steepest ascent for the function at the point (1, 1, 1).

(ii) The direction and rate of steepest descent for the function at the point (1, 1, 1).

(iii) The directional derivative of the function at the point (1, 1, 1) in the direction from the point (1, 1, 1) to the point (2, -1, -1).

[7]

(b) Find the tangent plane to the surface, described by the following equation:

$$x^2 + y^3 + z^4 = 3,$$

at the point (1, 1, 1). [5]

(c) Find and classify all the critical points of the following function:

$$f(x,y) = xy^2 - 3x^2 - y^2 + 2x + 2.$$

[7]

(d) Find the volume of the solid that is enclosed by the surfaces: $z = 0$, $z = y^2$, $y = x^2$ and $x = y^2$.

[6]

2. (a) Give the definition of the triple scalar product (also known as the mixed product). Prove that for any $\vec{a}$, $\vec{b}$ and $\vec{c}$,

$$\vec{a}\cdot(\vec{b}\times\vec{c}) = \vec{c}\cdot(\vec{a}\times\vec{b}) = \vec{b}\cdot(\vec{c}\times\vec{a})\ .$$

[6]

(b) Consider the curve C given by the following parametrization:

$$x = \sin^2(t), \quad y = 1-\cos(t), \quad z = 2t.$$

Find the unit tangent vector $\vec{T}$ to the curve C at point M_0 corresponding to $t = t_0 = \dfrac{\pi}{2}$. [6]

(c) Consider a vector field $\vec{F}$ defined on the whole 3-dimensional space and let $\vec{G} = c\vec{F}$, where $c \in \mathbb{R}$ is a constant. Prove, showing all steps, the following statement: if $\vec{F}$ is a conservative vector field, then the field $\vec{G}$ is also conservative. [5]

(d) Consider a counterclockwise-oriented closed contour C consisting of two parts C_1 and C_2. The curve C_1 is defined by its parametrization as $C_1 : x(t) = R\cos(\pi t), y(t) = R\sin(\pi t)$ for $0 \le t \le 1$ and the curve C_2 is defined as $C_2 : x(t) = Rt, y(t) = 0$ for $-1 \le t \le 1$, where $R \in \mathbb{R}$ and $R > 0$. Compute the line integral $\oint_C \vec{F}\cdot \mathrm{d}\vec{r}$ of the vector field $\vec{F} = \left(\cos x - \dfrac{y^3}{3}, \dfrac{x^3}{3} + \sin y\right)$ along the contour C. [8]

Section B

3. Consider the triple integral as follows:

$$J = \iiint\limits_{T} xy dx dy dz,$$

where the domain T is the tetrahedron with its vertices at (0, 0, 0), (2, 0, 0), (2, 2, 0) and (2, 0, 2).

(a) Express the integral as repeated integrals using

(i) Cartesian coordinates,

(ii) Cylindrical coordinates, and

(iii) Spherical coordinates.

[18]

(b) Calculate the integral using any one of the three coordinates, using the result obtained in (a), or otherwise. [3]

(c) Using the result of (b), or otherwise, calculate the following integral:

$$K = \iiint\limits_{S} \left(xy + xy^2 \sinh(z)\right) dx dy dz,$$

where the domain S is the union of the domain T and its image in the Oxy plane. [4]

4.

(a) Consider a function $f(x,y,z) : \mathbb{R}^3 \to \mathbb{R}$, where $f(x,y,z)$ is twice differentiable in a spatial domain $D \subset \mathbb{R}^3$. The Laplace equation written for the function $f(x,y,z)$ in the domain D is

$$\Delta f \equiv \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0.$$

(i) Demonstrate, showing all steps, that $\Delta f = $ div(grad(f)).

(ii) Let $f(x,y,z) \equiv f(r) = \dfrac{1}{r}$, where $r = \sqrt{x^2+y^2+z^2}$, and let the domain D be defined as $D : x > 0, y > 0, z > 0$. Is the function $f(r)$ a solution to the Laplace equation in the domain D? Explain your answer, showing all steps.

[12]

(b) Consider the vector field

$$\vec{F} = \left(\left(\frac{x^3}{5} + \frac{z^2 y^4}{10} + \frac{z^5}{15} \right), \left(\frac{x^5 z}{25} + \frac{y^3}{5} + \frac{x^3 z^2}{10} \right), \left(\frac{y x^5}{10} + \frac{x^5 y^3}{15} + \frac{z^3}{5} \right) \right)$$

and the spherical shell Ω defined as Ω: $a^2 \leq x^2 + y^2 + z^2 \leq b^2$, where $a, b \in \mathbb{R}$ and $0 < a < b$. Find the surface integral $\int \vec{F} \cdot \vec{dS}$ over the surface of the spherical shell Ω.

[13]

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LI/LH Multivariable & Vector Analysis

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats and outer-wear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) **MUST** be placed in the designated area.
- Check that you **DO NOT** have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **MUST** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are **NOT** permitted to use a mobile phone as a clock. If you have difficulty in seeing a clock, please alert an Invigilator.
- You are **NOT** permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately.
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to the Student Conduct procedures.